NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2 QUESTIONS AND SOLUTIONS.

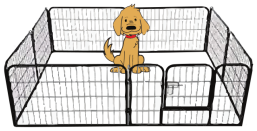
SECTION A : 5 marks each

1. Mary was born on January 1, 2000. Her mother was born on January 1, 1985 and her father was born on January 1, 1980. In what year was the sum of their ages 104?

Solution

Let x be the year in which the sum of their ages is 104. Mary's age is $x - 2000$, her mother's age is $x - 1985$ and her father's age is $x - 1980$. Now, summing their ages gives $(x - 2000) + (x - 1985) + (x - 1980) = 104$ hence $x = 2023$. It is/was the year 2023.

2. Mr. Otim's dog outgrew its rectangular pen that measured 3 metres by 5 metres. To make it



larger, he increased each side by an equal amount, which increased the pen's area by 9 square metres. What is the greater side length of the larger pen?

Solution

Let x be the amount of side length by which Mr. Otim increased the pen. It then follows that the new area of the rectangular pen is $(3 + x)(5 + x) = 3 \times 5 + 9$, hence $x = 1$ metre long. Therefore, the greater side length of the larger pen is $5 + 1 = 6$ metres.
Remark: Expanding $(3+x)(5+x) = 3\times5+9$ results in a quadratic equation given as $x^2+8x-9 = 0$ with solutions $x = -9$ and $x = 1$. We take a positive answer since we cannot have a negative length (-9) being added.

3. How many ways are there to arrange the four integers 1, 2, 3 and 4 in a row so that no two adjacent numbers have a sum of 5?

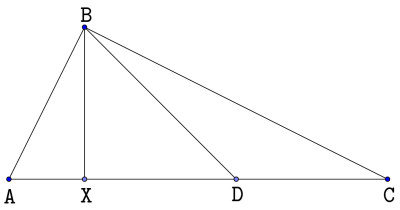
Solution

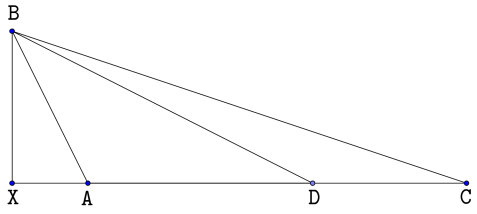
There are 4 ways of selecting the first term of the four integers. However for each first term, there are 2 ways of selecting the second term since $5 = 2 + 3 = 3 + 2 = 1 + 4 = 4 + 1$. Now, there is 1 way of selecting the third term and 1 way of selecting the fourth term. By the product rule, there are $4 \times 2 \times 1 \times 1 = 8$ ways to arrange the four integers 1, 2, 3 and 4 in a row so that no two adjacent numbers have a sum of 5. Alternatively, accept the list $[1, 2, 4, 3], [1, 3, 4, 2], [2, 1, 3, 4], [2, 4, 3, 1], [3, 1, 2, 4], [3, 4, 2, 1], [4, 2, 1, 3], [4, 3, 1, 2]$
Remark: A candidate can list all the $4! = 24$ ways of arranging the four integers and then picks out those such that no two adjacent numbers have a sum of 5.

4. A triangle ABC has an area of 40 units^2. A point D drawn on side AC such that AD:DC = 3:2. What is the area of triangle BDC?

Solution

Let X be the foot of the perpendicular draw from B to the line $\overrightarrow{AC}$ as indicated in any of the sketches below (with $\overrightarrow{AC}$ regarded as the base). The area of triangle ABC is $\frac{1}{2} \times |\overrightarrow{AC}||\overrightarrow{BX}| = 40$ units^2, hence $|\overrightarrow{AC}||\overrightarrow{BX}| = 80$ units^2. From the ratio $\overrightarrow{AD} : \overrightarrow{DC} = 3 : 2$, we have $|\overrightarrow{DC}| = \frac{2}{5}|\overrightarrow{AC}|$. The area of triangle BDC is $\frac{1}{2} \times |\overrightarrow{DC}||\overrightarrow{BX}| = \frac{1}{2} \times \frac{2}{5} \times |\overrightarrow{AC}||\overrightarrow{BX}| = \frac{1}{2} \times \frac{2}{5} \times 80 = 16$ units^2.





Alternatively, let the area of triangle BDC be α and length of $\overrightarrow{BX}$ be y. From the theory of ratios and areas of triangles, we have $\frac{\alpha}{40} = \frac{|\overrightarrow{CD}| \times y}{|\overrightarrow{AC}| \times y} = \frac{\frac{2}{3+2} \times |\overrightarrow{AC}|}{|\overrightarrow{AC}|} = \frac{2}{5}$, hence $\alpha = 16$ units^2.

5. Suppose x & y are positive real numbers and a & b are integers such that $(x^{3b+4}y^{3a-2})^4 = x^{b-6}y^{a+3}$, determine the value of $a - b$.

Solution

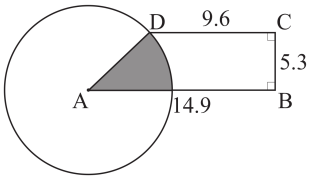
If a and b are integers, then so is $a - b$. From $(x^{3b+4}y^{3a-2})^4 = x^{b-6}y^{a+3}$,
we obtain, $x^{12b+16}y^{12a-8} = x^{b-6}y^{a+3}$. It follows that $12b + 16 = b - 6$ and $12a - 8 = a + 3$.
Hence $a = 1$ and $b = -2$, and this gives $a - b = 3$.

6. Everyday, Mr. Wafula walks to the market 4 km away at a speed of 3 km/h and then rides back home on a boda bike at twice this speed using a route that is 800 metres longer. How long does Mr. Wafula's journey to the market and then back home last (in minutes)?

Solution

Let t_1 be the time Mr. Wafula takes to go to the market and t_2 be time he takes to return to his home. Clearly, $t_1 = \frac{4}{3} \times 60 = 80$ minutes and $t_2 = \left(\frac{4 + 0.8}{2 \times 3}\right) \times 60 = 48$ minutes. Therefore, Mr. Wafula takes $t_1 + t_2 = 80 + 48 = 128$ minutes.

7. In trapezoid ABCD, below, vertex A is the center of the circle, vertex D lies on the circle, and vertices B and C are right angles. If AB = 14.9 cm, BC = 5.3 cm and DC = 9.6 cm, what is the area (as a decimal to the nearest tenth) of the shaded sector of the circle?



Solution

Drop an auxiliary line from D, perpendicular to AB, intersecting AB at E. Then EB = DC = 9.6 cm, so AE = AB - EB = (14.9 - 9.6) = 5.3 cm = BC = ED. Therefore, right triangle ADE is isosceles. So, angle DAE = 45°, meaning the sector is $\frac{1}{8}$ of the circle. By properties of 45-45-90 right triangles, circle centred at A has radius AD = $5.3\sqrt{2}$ cm, so sector DAF has area $\frac{\pi}{8} \times (5.3\sqrt{2})^2 \approx 22.1\text{cm}^2$. Alternatively, let x be the radius of the circle and θ be angle DAB. By Pythagora's theorem, $x = \sqrt{5.3^2 + (14.9 - 9.6)^2} = 5.3\sqrt{2}$ cm and $\theta = \arcsin\left(\frac{5.3}{x}\right) = \frac{\pi}{8}$ radians. The area of the shaded sector is $\theta x^2 = \frac{\pi}{8} \times (5.3\sqrt{2})^2 \approx 22.1$ cm^2.

8. The International Standard Book Number (ISBN) is a numeric commercial book identifier that is intended to be unique. It is usually a 13 digit number that is given to each new book which is published. If the sum of the 2nd, 4th, 6th, 8th, 10th and 12th digits is tripled and then added to the remaining digits (1st, 3rd, 5th, 7th, 9th, 11th and 13th), the total will always be divisible by 10. For example, in this case of the `ISBN 978-31-6148-410-0` given below, we have $3 \times (7+3+6+4+4+0) + (9+8+1+1+8+1+0) = 72 + 28 = 100$, which is divisible by 10. What digit is represented by # in the `ISBN 9#8-0-90-658898-7` if this `ISBN` satisfies the above property?



Solution

Let the second digit be x. Clearly, $x \in \{0, 1, \ldots, 9\}$. From the above conditions, we must have $3 \times (x+0+0+5+8+8) + (9+8+9+6+8+9+7) = 3(21+x) + 56 = 63 + 3x + 56 = 119 + 3x$. Now $119 + 3x$ is divisible by 10 if and only if $x = 7$.

9. A cylinder whose height is 5 times its radius is inscribed in a cone whose height is 7 times its radius. What fraction of the cone's volume lies inside the cylinder?

Solution

Let the cylinder's radius be x units and that of the cone be y units. Therefore, the cylinder height is $5x$ units while the cone's height is $7y$ units. By similar triangles, we have $\frac{y}{x} = \frac{7y}{7y-5x}$, hence $7y^2 = 12yx$. Since $y \neq 0$, it follows that $7y = 12x$. The volume of the cylinder is $\pi x^2(5x)^2$ while that of the cone is $\frac{1}{3}\pi y^2(7y)$. The fraction of the cone's volume lies inside the cylinder is $\frac{\pi x^2(5x)}{\frac{1}{3}\pi y^2(7y)} = \frac{15x^3}{7y^3} = \frac{3 \times 7^3}{7 \times 12^3} = \frac{245}{576}$ since $7y = 12x$.

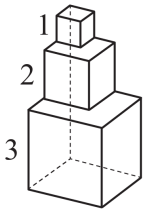
10. The operation $*$ is defined by $x * y = xy - x - y + 2$. If $7 * b = 13$, what is the value of $b * b$?

Solution

We have $13 = 7 * b = 7b - 7 - b + 2$ hence $b = 3$. So $b * b = 3 * 3 = 3 \times 3 - 3 - 3 + 2 = 5$.

SECTION B : 10 marks each

11. Three solid cubes of edge lengths 1 cm, 2 cm and 3 cm respectively are stacked together vertically with the largest at the bottom and the smallest at the top (along the same vertical axis). What is the surface area of the solid object formed?



Solution

The surface area of the solid is the sum of the surface areas of the three cubes minus twice the overlap area among adjacent cubes. [The "twice" is due to one face on each of two adjacent cubes being obscured.] The surface area of a cube of edge length s is $6s^2$, so for the three cubes totals $6(1^2 + 2^2 + 3^2) = 6 \times 14 = 84\text{cm}^2$. With each pair of adjacent cubes, the loss of surface area is twice the area of a face of the smaller cube, therefore totaling $2(1^2 + 2^2) = 2 \times 5 = 10\text{cm}^2$. The net surface area is $(84 - 10)\text{cm}^2 = 74\text{cm}^2$.

12. If two standard, fair 6-sided dice are rolled, what is the probability that the product of the two numbers rolled is a perfect square? Express your answer as a common fraction.

Solution

Assume rolling of the dice twice are two independent events. As such there are 36 possible outcomes, namely $\{(1,1),(1,2),\ldots,(6,5),(6,6)\}$. Of these, the desired outcomes are $\{(1,1),(1,4),(2,2),(4,1),(3,3),(4,4),(5,5),(6,6)\}$, (so that the product of the outcome is a perfect square less than or equal to 36). Therefore, the probability that the product of the two numbers rolled is a perfect square is $\frac{8}{36} = \frac{2}{9}$.

13. A satellite is an object in space that circles/orbits around a bigger object, for example the moon is an Earth bound satellite. A satellite, located 400 km above Earth's surface, travels at a speed of 32,000 km/h. For simplicity, assume that the Earth is a perfect sphere with circumference of any great circle of 40,000 km. How many minutes does it take the satellite to orbit Earth one time? Express your answer as a decimal to the nearest tenth.

Solution

Let x km be the radius of the Earth. Since we take the Earth to be a perfect sphere, we have $2\pi x = 40,000$. So the circumference of the satelitte's orbit is $2\pi\left(400 + \frac{40,000}{2\pi}\right)$. Since the satellite travels at $32,000$ km/h, it takes $\frac{2\pi\left(400+\frac{40,000}{2\pi}\right)}{32000} = (\frac{\pi}{40} + \frac{5}{4})\text{hours} = (\frac{\pi}{40} + \frac{5}{4}) \times 60\text{minutes} = 79.7\text{minutes}$ to complete one orbit around the Earth.

14. An hour ago, a subaru forester driver heading to Lira observed 78987 km on his odometer reading. This number is palindromic, meaning that it reads the same whether backwards or forwards. How fast was he driving during the last hour if the current number on his odometer is also palindromic?



Solution

Let x be the current number on the odometer. It is clear that $78987 \leqslant x \leqslant 79207$ since the speed limit of the car is 220 km/h (hence covers at most 220 km in one hour). The possible palindromes in this range are 78987 and 79197. Therefore, the subaru forester driver was driving at $\frac{(79197-78987)}{1} = 110\text{km/h}$.

15. A trime is a three-digit integer for which any two digits of the integer, ordered as they appear in the original integer, form a two-digit prime number. For example, 137 is a trime because 13, 17 and 37 are all two digit prime numbers. However, 211 is not a trime because 21 is not prime (much as 11 is a two digit prime). What is the greatest trime?

Solution

We can find the greatest trime by looking at the prime numbers between 1 and 100, which are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97. We can see from the provided example that in order to form a trime, we can "merge" two double-digit prime numbers that share a digit. This way, we know that at least two of the two-digit numbers produced are prime, and simply need to check the third. Since we are looking for the greatest trime, we can start with 97 and work our way down. 97 can be "merged" with any of the primes in the 70s, so 79, 73, or 71. However, 979 produces 99 as one of its two-digit numbers, 973 produces 93 as one of its two-digit numbers, and 971 produces 91 as one of its two-digit numbers, all of

which are not prime. 89 can be "merged" with 97 to make 897, but 87 is not prime. 83 can be merged with 37 or 31, but neither of 87 or 81 are prime. 79 can be "merged" with 97 to make 797, which produces 77 as one of its two-digit numbers, which is not prime. 73 can be "merged" with 37 or 31. While 737 produces 77 as one of its two-digit numbers, which is not prime, 731 produces three two-digit primes: 73, 31, and 71. So, 731 is the greatest possible trime.

END